# PSH Exhaustions

## Threefold counterexamples and surface criteria

One manuscript • Current and previous versions • Revision record

This collection examines the distinction between continuous and smooth plurisubharmonic (psh) exhaustions. It packages a revised manuscript from the user's *vibemath* research work, together with its earlier version and supplied revision materials.

**Current paper.** *Continuous and Smooth Plurisubharmonic Exhaustions: Threefold Counterexamples and Partial Results for Surfaces.* The supplied revision has 27 pages; the archived predecessor has 47. Both source/PDF pairs are preserved.

### A. A Kähler threefold with a regularity gap

Theorem 1.1 constructs a smooth quasi-projective Kähler threefold

$$M = \mathrm{Tot}(L^{-1})$$

admitting a locally Lipschitz psh exhaustion but no $C^2$ psh exhaustion. The compact base is a ruled surface from Koike's modified Grauert example. The construction combines a continuous semipositive metric with an obstruction to every $C^2$ exhaustion.

### B. The same gap on a non-Kähler threefold

Theorem 1.2 constructs, over any point $x \in M$, a proper modification $\mu : N \to M$ that is biholomorphic away from the fiber over $x$. The smooth threefold $N$ is non-Kähler and admits a locally Lipschitz psh exhaustion but no $C^2$ one. Hironaka's opposite-order blowups are an attributed input.

### C. A line-bundle radius criterion

For a holomorphic line bundle $A$ over a compact complex manifold, Corollary 3.4 gives, for $r = 2$ and $r = \infty$,

$$\mathrm{Tot}(A^{-1}) \text{ has a } C^r \text{ psh exhaustion}$$
$$\Longleftrightarrow$$
$$A \text{ has a } C^r \text{ semipositive Hermitian metric.}$$

Circle averaging and a high-level radius function give the reverse direction to the usual squared-norm construction.

### D. Partial affirmative results for surfaces

Theorem 5.4 treats complements of nonzero effective divisors in nonprojective compact surfaces. Theorem 5.5 treats smooth $\mathbf{P}^1$-bundles with nonempty reduced simple normal crossing boundary. In both settings, continuous and smooth psh exhaustibility is equivalent to the absence of a negative definite connected boundary component.

# Scope, dependencies, and reading guide

## What the surface section establishes

Proposition 5.2 shows that for a line bundle $A$ over a compact Riemann surface, continuous exhaustibility of $\mathrm{Tot}(A^{-1})$, nonnegative degree of $A$, and smooth exhaustibility are equivalent.

Proposition 5.6 gives further projective sufficient cases; Proposition 5.8 gives smooth square-zero boundary criteria. Corollary 5.7 supplies necessary numerical conditions for projective pairs remaining after these reductions. Those conditions do not construct a surface counterexample.

Table 1 concerns compact *minimal* surface compactifiers under its explicit hypotheses. A change of compactifier can change the open complement. The table is not a classification of arbitrary noncompact surfaces.

## Proof map and revision history

**Sections 2–3.** A cusp correction near a negative curve reaches the locally Lipschitz endpoint. Global gluing produces a metric; the radius criterion transfers the known $C^2$ metric obstruction to an exhaustion obstruction.

**Section 4.** Compactly supported modifications preserve the relevant exhaustion properties. A localized Hironaka configuration supplies the non-Kähler example.

**Section 5.** Boundary intersection matrices, collar constructions, surface classification, abundance, and neighborhood results yield the stated affirmative cases.

Sections 2–4 and the two principal theorem statements are unchanged from the predecessor. Source hashes, the supplied patch, and the Chinese change report document the revision.

## Attribution and verification status

The ruled-surface obstruction is credited to earlier work. Supporting criteria are not uniformly claimed as new; priority checking remains qualified.

Some surface branches invoke Theorem 1.6 of the cited September 2026 Chen–Filip–Sun–Tosatti–Zhang preprint. The supplied report records incomplete access to Vâjâitu's published full text. These inherited qualifications remain in the repository.

Repository checks cover integrity, links, references, compilation, and layout. External journal review and formal verification are not established. Authorship and licensing remain unassigned.

## Questions left unresolved by this manuscript

The general continuous-to-$C^2$ exhaustion problem in complex dimension two is not resolved. Neither minimal counterexample dimension nor the optimal regularity threshold is determined. In particular, existence of a $C^1$ exhaustion on the constructed threefolds is not decided here.

## Repository guide

Start with `CONTENTS.md` for precise statements and locations. The current paper is in `preprints/`; the predecessor is in `archive/`. See `reviews/` for scope notes and revision materials, `reasoning_traces/` for the editorial summary, and `UPLOAD_TO_GITHUB.md` for publishing instructions.

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